

Smart Fuel Management and Eco-Driving Assistant

Complete Feature Set for Malaysian Drivers

0. User & Vehicle Profile Module

Purpose

Store user and vehicle information as the foundation for all personalized analytics, cost estimation, and environmental tracking across the platform.

0.1 User Account Management

Users can register, login, logout, and update their profile. Information stored includes name, email, optional phone number, and optional profile picture.

0.2 Vehicle Profile Management

Users register their vehicle with the following details: vehicle brand (Perodua, Proton, Honda, Toyota, etc.), vehicle model (Myvi, Bezza, City, Vios, etc.), manufacturing year, fuel type (RON95, RON97, Diesel), engine capacity (1.3L, 1.5L, 2.0L, etc.), fuel tank capacity in litres, and transmission type (Automatic or Manual).

0.3 Multi-Vehicle Support

Users can register and manage more than one vehicle under a single account. Each vehicle maintains its own fuel logs, efficiency history, eco score, and carbon emission records. Users can switch between vehicles from the dashboard.

0.4 Vehicle Fuel Efficiency Baseline

The system stores an estimated fuel efficiency benchmark based on the selected vehicle model as a reference before enough user data is collected and as a fallback when fuel logs are incomplete.

Examples:

- Myvi 1.5 — 5.5 L/100km
- Bezza 1.3 — 4.8 L/100km

0.5 Monthly Fuel Budget Setup

Users define a weekly or monthly fuel budget in Ringgit Malaysia. The system tracks actual spending against this budget and triggers alerts when thresholds are approached or exceeded.

0.6 Environmental Goals

Users set personal sustainability targets such as reducing fuel consumption by 10%, improving their Eco Score above 80, or reducing monthly carbon emissions by a specific amount. Progress is tracked and reported on the dashboard.

1. Fuel Price Intelligence Module

Purpose

Help users make informed refuelling decisions by monitoring Malaysian fuel price changes and calculating their direct impact on personal transportation costs.

1.1 Current Fuel Price Dashboard

Displays the latest government-regulated prices for RON95, RON97, and Diesel. Prices are updated automatically each time the Malaysian government announces a revision.

1.2 Fuel Price History Analysis

Visualizes fuel price movements through weekly, monthly, and yearly trend charts so users can identify periods of price increase or decrease and anticipate future changes.

1.3 Fuel Cost Impact Calculator

Allows users to estimate the cost effect of price changes based on their personal driving data. Given a monthly travel distance and vehicle fuel efficiency, the system calculates monthly fuel consumption, monthly fuel expenses, and the cost difference resulting from any price adjustment.

1.4 Fuel Price Change Notifications

Automatically alerts users when fuel prices increase or decrease, or when a new weekly price is announced. Notifications include the estimated additional or saved monthly cost based on the user's actual consumption pattern.

Example: "RON97 increased by RM0.10/L this week. Estimated additional monthly cost: RM18."

1.5 Nearby Fuel Station Finder

Displays fuel stations near the user's current location or along a planned route. Stations are shown on a map with their distance, available fuel types, and where data is available, user-submitted price or promotion information. This supports smarter refuelling timing and location decisions.

2. Smart Fuel Consumption Tracker

Purpose

Provide complete visibility into fuel usage, cost, and driving efficiency using actual refuelling records, OCR receipt scanning, and trip-level estimation.

2.1 Manual Fuel Logging

Users manually enter the refuelling date, fuel type, litres purchased, total amount paid, and odometer reading. All records are stored and used for consumption calculations and trend analysis.

2.2 Automatic Fuel Logging via OCR

Users upload a fuel receipt image or e-receipt screenshot. The system uses optical character recognition to extract the date, fuel station name, litres purchased, and total amount paid. This reduces manual entry errors and improves data completeness.

2.3 Actual Fuel Consumption Calculation

The system computes real fuel efficiency using two consecutive refuelling records.

Formula: Fuel Efficiency (km/L) = Distance Travelled / Fuel Used

Example: Previous odometer 10,000 km, current odometer 10,450 km, fuel filled 35 L.
Distance = 450 km. Efficiency = $450 / 35 = 12.86$ km/L.

This method produces the ground truth consumption figure used across all analytics.

2.4 Trip-Level Fuel Consumption Estimation

Estimates fuel usage for a specific trip before or after refuelling, using trip distance from GPS and the user's historical average efficiency or the vehicle baseline as a fallback.

Formula: Estimated Fuel (L) = Distance (km) / Average Efficiency (km/L)

This supports route optimization, trip cost estimation, and per-trip carbon emission calculation.

2.5 Fuel Efficiency Trend Analysis

Tracks and displays the user's current km/L trend, best recorded efficiency, worst recorded efficiency, and efficiency variability over time. Insight examples include identifying improvement or decline trends and flagging inconsistent driving behaviour.

2.6 Monthly Fuel Spending Analysis

Tracks total monthly fuel cost, weekly spending breakdowns, and spending trend comparisons against prior months. Insight example: "Fuel expenses increased by RM45 (+12%) compared to last month."

2.7 Vehicle Performance Monitoring

Detects anomalies in fuel usage patterns that may indicate driving inefficiency, changes in traffic conditions, or potential vehicle maintenance issues. Triggered signals include a sudden drop in km/L, increased fuel cost per km, or irregular consumption patterns. Example insight: "Fuel efficiency dropped 18% — possible tyre pressure issue or driving pattern change."

2.8 Maintenance Reminder System

Based on mileage intervals and the anomalies detected in Module 2.7, the system reminds users to check or service items that directly affect fuel efficiency. Reminders cover tyre pressure checks, engine oil changes, air filter replacement, and tyre rotation. Users can also manually set custom service reminders by mileage or date.

3. Driving Behaviour Analytics Module

Purpose

Help users understand how their driving habits affect fuel consumption and costs, and guide them toward more efficient driving patterns.

3.1 Driving Style Monitoring

Using GPS and smartphone sensors, the system detects and records the following behaviours:

Aggressive acceleration — rapid speed increase from low speed, associated with higher fuel consumption.

Harsh braking — frequent sudden braking indicating poor anticipation and inefficient driving.

Excessive speeding — driving significantly above optimal fuel efficiency speeds.

Long idling — engine running while the vehicle remains stationary, such as in parking lots or at roadsides.

3.2 Eco-Driving Score

Generates a score from 0 to 100 based on four weighted factors.

- Acceleration: 25%
- Braking: 25%
- Idling: 20%
- Speed consistency: 30%

Score interpretation: 90 to 100 is Excellent, 75 to 89 is Good, 50 to 74 is Moderate, below 50 is Poor.

3.3 Personalized Driving Recommendations

Delivers specific, data-driven suggestions based on detected behaviour patterns. Examples: "Reduce aggressive acceleration to improve fuel efficiency by approximately 10%." and "You spent 42 minutes idling this week, consuming an estimated 0.8 litres of fuel."

4. Carbon Emission Monitoring Module

Purpose

Increase individual awareness of personal transport-related environmental impact and support users in reducing their carbon footprint.

4.1 Carbon Emission Calculation

Estimates CO₂ emissions from fuel consumption using established conversion factors.

Formula: CO₂ (kg) = Fuel Consumed (L) × 2.31 (petrol) or × 2.68 (diesel)

This calculation runs automatically for every recorded refuel and every estimated trip, feeding all downstream emission displays and comparisons.

4.2 Trip-Level Emissions Display

For every recorded or estimated trip, displays the distance travelled, fuel used, and CO₂ emitted. This gives users immediate visibility of the environmental cost of each individual journey and allows comparison across trips of similar routes.

4.3 Eco-Driving Recommendation Engine

Purpose

Act as the platform's unified solution engine for environmental improvement. Rather than scattering tips across modules, this engine cross-references all available data sources and generates coordinated, prioritised, quantified recommendations telling the user exactly what to change and what the measurable impact will be.

Recommendations are generated from three data sources:

Driving behaviour data

- Harsh acceleration detected → recommendation to ease off throttle with estimated fuel saving
- Hard braking detected → recommendation to increase following distance and anticipate stops
- Excessive idling detected → recommendation to switch off engine after a defined threshold

Route data

- Frequently congested route identified → recommendation of a less congested alternative with estimated emission reduction per trip

- Route history showing consistent high fuel burn → flag for route review

Fuel efficiency trends

- Declining km/L over recent weeks → warning with recommended actions such as tyre pressure check or driving style adjustment
- Efficiency below vehicle baseline → prompt to review both driving habits and vehicle condition

Example recommendations:

- "Reduce rapid acceleration to save up to 10% fuel consumption."
- "Avoid idling for more than 2 minutes to reduce CO₂ emissions."
- "Route B reduces emissions by 0.8 kg CO₂ per trip compared to your usual route."
- "Maintaining steady speed between 80 and 90 km/h improves fuel efficiency by 12 to 15%."

Recommendations are ranked by estimated impact so users address the highest-gain changes first. Each recommendation links to the specific data that triggered it so users understand the evidence behind the suggestion.

4.4 Eco-Score (Environmental Rating)

Purpose

Convert overall environmental performance into a single, easy-to-understand score that reflects not just driving style but the full picture of a user's transport-related emissions.

Range: 0 to 100

Scoring factors:

- Fuel efficiency: 40%
- Driving behaviour: 30%
- Route efficiency: 30%

Score interpretation:

- 90 to 100 — Excellent, low emissions
- 70 to 89 — Good
- 50 to 69 — Moderate
- Below 50 — High environmental impact

This score is distinct from the Eco-Driving Score in Module 3.2. A user can drive smoothly and score well on driving behaviour but still carry a lower Eco-Score if they consistently take congested routes or drive a fuel-heavy vehicle. The Eco-Score captures the complete environmental outcome, not just the driving inputs.

The score updates after each trip and is prominently displayed on the unified dashboard so users always know their current environmental standing.

4.5 Monthly Environmental Dashboard

Shows total CO₂ emissions for the current month, weekly emission trends, and month-over-month comparison. Users can view progress toward the environmental goals they set in Module 0.6, displayed as a percentage of target achieved alongside the absolute figures.

4.6 Sustainability Insights

Delivers contextual insights comparing current emission performance to prior periods, benchmark figures, and user-defined goals. Examples: "Your eco-driving improvements reduced emissions by 12% this month." and "You prevented approximately 8 kg of CO₂ emissions compared to last month."

Where sufficient data exists, the system projects the user's estimated annual CO₂ output based on current patterns, giving a longer-term perspective on their environmental impact.

4.7 Alternative Transport Mode Comparison

For selected trips or routes, estimates and displays the approximate fuel cost and CO₂ emissions of the same journey made by public transport, carpool, or other available modes where relevant. This broadens the environmental perspective beyond personal vehicle optimisation and gives users a concrete basis for deciding when an alternative is worth considering.

5. Route Optimization Module

Purpose

Reduce travel cost and fuel consumption through intelligent route selection that considers distance, traffic, fuel use, and cost — not just travel time.

5.1 GPS Navigation Integration

Users enter a start location and destination. The system retrieves available routes using integrated mapping services.

5.2 Fuel-Efficient Route Recommendation

Evaluates routes not only by distance and time but also by estimated fuel consumption and fuel cost. Recommends the route that offers the best balance of efficiency and practicality for the user.

5.3 Route Comparison Dashboard

Presents multiple route options side by side with distance, estimated travel time, estimated fuel consumption, and estimated cost so the user can make an informed choice.

Example:

- Route A — 18 km, 30 min, 1.5 L, RM3.15
- Route B — 20 km, 24 min, 1.2 L, RM2.52
- System recommendation: "Route B saves approximately RM0.63 in fuel."

5.4 Traffic-Aware Fuel Prediction

Uses real-time traffic data to adjust fuel consumption and cost estimates. Example: "Heavy congestion on this route may increase fuel consumption by 18%."

5.5 Route History and Learning

Stores the user's frequently travelled routes and learns average fuel usage, travel time, and cost per route over time. This data continuously improves the accuracy of future recommendations and supports commute planning.

5.6 Pre-Trip Journey Planner

Before departing, users can simulate a planned journey to see the estimated total fuel consumption, cost, and CO₂ emissions across route options. This helps users decide when to refuel, which route to take, and whether an alternative transport mode is worth considering for that specific trip.

6. Smart Refuelling Decision Assistant

Purpose

Convert real-time and historical data into clear, timely refuelling recommendations so users never overspend, under-plan, or miss a price opportunity.

6.1 Refuelling Recommendations

Combines current fuel level, upcoming price change schedule, user's driving schedule, and average daily consumption to generate actionable refuelling advice. Examples: "Current fuel is sufficient for approximately 4 days of your typical driving." and "Based on your pattern, refuelling 25 litres now is sufficient until next week's price update."

7. Unified Analytics Dashboard

Purpose

Centralize all transportation data and insights in one accessible view so users never need to navigate multiple screens to understand their overall fuel and driving performance.

7.1 Financial Analytics

Displays total fuel spending, monthly spending trends, and cost per kilometre.

7.2 Fuel Analytics

Shows current fuel efficiency, consumption patterns over time, and efficiency trends across vehicles.

7.3 Driving Analytics

Presents the Eco-Driving Score, driving behaviour event trends, and week-over-week comparisons.

7.4 Environmental Analytics

Summarizes total carbon emissions, emission trends, and progress toward user-defined sustainability goals.

7.5 Route Analytics

Highlights frequently used routes, fuel saved through route optimization, and efficient route statistics over time.

8. Monthly Report Generation Module

Purpose

Give users a consolidated, downloadable summary of their fuel management performance each month so all fuel expenses, driving behaviour, consumption data, and environmental impact are available in one document rather than fragmented across app screens.

8.1 Monthly Report Content

Each report covers the following sections: total fuel spending and comparison to budget, fuel efficiency trends and best and worst recorded km/L, Eco-Driving Score summary and top driving behaviour findings, total CO₂ emissions and comparison to prior month and sustainability goals, most frequently used routes and fuel saved through route optimization, maintenance reminders triggered during the month, and a summary of any fuel price changes and their estimated cost impact on the user.

8.2 Report Format and Export

Reports are generated in PDF format and available for download directly from the app. Users can also opt to receive the monthly report automatically by email at the end of each calendar month.

8.3 Report History

Users can access and download reports from previous months within the app for personal record-keeping, expense tracking, or employer reimbursement purposes.

9. Smart Notification and Alert System

Purpose

Encourage continuous user engagement and support proactive behaviour improvement through timely, relevant, and non-intrusive alerts.

9.1 Notification Types

Cost alerts — "Fuel spending has exceeded your monthly budget by 10%."

Efficiency alerts — "Fuel efficiency decreased by 8% this week."

Driving behaviour alerts — "Frequent harsh braking detected this week."

Route alerts — "A more fuel-efficient route is available for your usual commute."

Environmental alerts — "You reduced carbon emissions by 6% this week."

Refuelling alerts — "Fuel price expected to change this week. Consider refuelling before the update."

Maintenance alerts — "Your vehicle is due for a tyre pressure check based on recent mileage."

Price alerts — "RON95 price decreased by RM0.05/L effective today."

9.2 Notification Preferences

Users can configure which notification types they receive, set quiet hours, and choose between push notifications and email alerts.

Summary of Modules

- Module 0 — User and Vehicle Profile
- Module 1 — Fuel Price Intelligence
- Module 2 — Smart Fuel Consumption Tracker
- Module 3 — Driving Behaviour Analytics

- Module 4 — Carbon Emission Monitoring
- Module 5 — Route Optimization
- Module 6 — Smart Refuelling Decision Assistant
- Module 7 — Unified Analytics Dashboard
- Module 8 — Monthly Report Generation
- Module 9 — Smart Notification and Alert System